



Semester II Examinations 2008/ 2009

Exam Code(s) 1MF1, 1OA1, 1SD1, 3BS4, 3BS9, 3EL2, 4CS2, 4BS3

Exam(s) 3rd & 4th Science, H. Dip. Appl. Sc, M.Sc.
Software Design & Development

Module Code(s) MP307
Module(s) Modelling II

Paper No. 1
Repeat Paper

External Examiner(s) Professor D.C.G. McKeon
Internal Examiner(s) Dr P.T. Piironen
Professor T. Sherry

Instructions: For full marks answer question 1 in Section A **and**
TWO questions in Section B correctly.

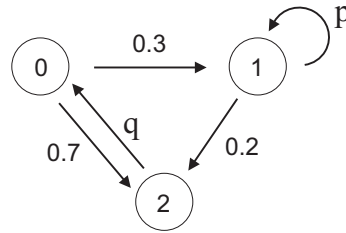
Duration 2hrs
No. of Pages 4
Department(s) Applied Mathematics
Course Co-ordinator(s) Dr P. Piironen

Requirements:

MCQ
Handout
Statistical/ Log Tables Yes
Cambridge Tables
Graph Paper
Log Graph Paper
Other Materials

Section A - Answer all questions

1. (a) Describe what is meant by a finite ergodic Markov process with transition probability matrix P . Define the equilibrium probabilities Π_i for a finite ergodic system and show that they obey the matrix equation $\Pi = \Pi P$. Describe a sufficient condition for P which guarantees ergodicity.
- (b) Consider the following Markov Process.



- (i) Find the transition matrix P and the probabilities p and q .
 - (ii) Is the Markov process ergodic?
 - (iii) Find the equilibrium probabilities.
- (c) Consider an ecological system with a prey population of size $x(t)$ and a predator population of size $y(t)$ which satisfy the differential equations

$$\begin{aligned}\frac{dx}{dt} &= x(a - by), \\ \frac{dy}{dt} &= y(-c + bx).\end{aligned}$$

- (i) Give an interpretation for the various terms in these equations.
 - (ii) Locate all equilibrium points.
- (d) Consider a linear control system

$$\begin{aligned}\frac{dx_1}{dt} &= a_1x_1 + a_2x_2 + b_1u, \\ \frac{dx_2}{dt} &= a_3x_1 + a_4x_2 + b_2u \\ y &= c_1x_1 + c_2x_2\end{aligned}$$

where x_1 and x_2 represent the state, $u \in \mathbb{R}$ is the input, $y \in \mathbb{R}$ is the output, and $a_1, a_2, a_3, a_4, b_1, b_2, c_1, c_2$ are real constants.

- (i) Describe what is meant by *observability* for this system.
- (ii) Describe what is meant by *reachability* for this system.

Section B - Answer any **two** questions.

2. (a) Consider a telephone system exchange with r operators. Assuming the probability of a call arriving is independent of the queue size, find the equilibrium probabilities.
- (b) An exchange consists of two operators each of which can deal with 10 customer enquiries per 15 minutes on average. If 160 calls per hour arrive on average what is the equilibrium probability for all lines to be busy? What would be the effect of introducing a third operator?
- (c) On a sunny day, on average, an ice cream salesman services 2 customers every minute with 40 customers arriving every hour. Using a Poisson model for the arrival and servicing mechanism, find the equilibrium probability that more than 2 customers are queuing. What is the average size of the queue? How would the queue change if 80 customers arrived per hour?
3. Two competing species with population sizes P_1, P_2 evolve according to the following differential equations

$$\begin{aligned}\frac{dP_1}{dt} &= 125P_1 - 5P_1^2 - 5P_1P_2, \\ \frac{dP_2}{dt} &= 100P_2 - 5P_1P_2 - 5P_2^2.\end{aligned}$$

- (a) Give an interpretation for the various terms in these equations.
- (b) Find the equilibrium points for P_1, P_2 .
- (c) Show that for any initial non-zero populations $P_1(t) \rightarrow 25$ and $P_2(t) \rightarrow 0$, as $t \rightarrow \infty$. What is the interpretation of this result?

4. A two-compartment model for drug delivery can be modelled as

$$\begin{pmatrix} \dot{c}_1 \\ \dot{c}_2 \end{pmatrix} = \begin{pmatrix} -p_0 - p_1 & p_1 \\ p_2 & -p_2 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} + \begin{pmatrix} b_0 \\ 0 \end{pmatrix} u, \quad y = c_2,$$

where c_1 and c_2 are the concentrations of the drug in each compartment, p_0 , p_1 , p_2 and b_0 are rate constants, u is the flow rate of the drug into compartment 1 and y is the concentration of the drug in compartment 2.

- (a) Draw a schematic diagram of this system including concentrations, flow directions and rate parameters.
- (b) Investigate for what values of p_0, p_1 and p_2 this system is observable.
- (c) Design a control function

$$u = -Kc + k_r r, \quad \text{for } K = \begin{pmatrix} k_1 & k_2 \end{pmatrix}, \quad c = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix},$$

where k_1, k_2, k_r are the control parameters, such that the desired output $r = 1$ and the closed loop eigenvalues for the system satisfies $\lambda_1 = -1$ and $\lambda_2 = -2$.